BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Sound, Sampling and File Sizes

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

SILENT

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: what does colour depth measure?

2. What is the relationship between bits and bytes?

3. Find the bits in a 10 x 5 image with 4-bit colour depth.

4. Why may an image file exceed its raw pixel size?

Today's objectives

Explain how sound is sampled and calculate uncompressed audio file sizes.

Key vocabulary - Read, cover, recall

sample rate Number of samples taken each second	**bit depth** Bits used to record one sample
duration How long the recording lasts	**analogue** Continuous, changing signal

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Sound becomes digital samples

A microphone receives analogue sound. A computer takes regular samples.

Sample rate = readings each second. Bit depth = bits per reading.

Digital readings of a changing sound signal



More samples per second record the changing signal in more detail.

I Do - Calculate an uncompressed file size

LISTEN

For MONO audio, calculate the raw number of bits first.

RATE x DURATION x BIT DEPTH = BITS

8000 x 2 x 8 = 128000 bits

128000 / 8 = 16000 bytes

Stereo adds a second channel, so raw size is twice as large.

We Do - Guided practice

MWBs

8,000 samples/sec x 3 seconds x 8 bits/sample: calculate bytes.

Samples: _____ x _____ = _____

Bits: _____ x _____ = _____

Bytes: _____ / 8 = _____

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Think and write

Code First - Predict > Write > Test > Improve

MWBs

Start on paper. Do not look at model answers until page 4.

```
rate = 8000    duration = 2    depth = 8
print(rate * duration * depth)
```

Predict the number of bits printed by the code.

Change the code to display bytes instead of bits.

Improve the code to calculate bytes for 3 seconds.

You Do - Independent work

SILENT

Define sample rate and bit depth. [2]

8,000 Hz for 2 seconds at 8 bits: calculate bytes. [3]

16,000 Hz for 3 seconds at 16 bits: calculate bytes. [3]

Thinking: What changes if the sound is recorded in stereo?

Exit Ticket - True or False?

THUMBS

Answer before you turn to the green-pen page.

1. A higher sample rate records more samples each second. ☐ T ☐ F
2. Bit depth is measured in seconds. ☐ T ☐ F
3. A longer recording needs more raw data at the same settings. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 Bits used for each pixel. ☐

Q2 8 bits make one byte. ☐

Q3 10 x 5 x 4 = 200 bits. ☐

Q4 Headers and metadata add data. ☐

We Do - Guided practice

8000 x 3 x 8 / 8 = 24000 bytes

Key vocabulary

sample rate ☐
Number of samples taken each second

bit depth ☐
Bits used to record one sample

duration ☐
How long the recording lasts

analogue ☐
Continuous, changing signal

Code First - Answers

GREEN PEN

Predict ☐

8000 x 2 x 8 = 128000 bits

Write ☐

print(rate * duration * depth / 8) = 16000

MWBs

Improve ☐

Set duration = 3: 24000 bytes

Exit Ticket - Explanations

1. True

2. False: bit depth is bits per sample

3. True

You Do - Self-mark

GREEN PEN

Q1	Rate = samples/second; depth = bits/sample	☐
Q2	8000 x 2 x 8 / 8 = 16000 bytes	☐
Q3	16000 x 3 x 16 / 8 = 96000 bytes	☐
Thinking	Stereo doubles the size if both channels use the same settings.	☐

Try the question again

GREEN PEN

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: